

## 5

# Big Data Analytics and Model Evaluation

## Syllabus

**Clustering Algorithms :** K-Means, Hierarchical Clustering, Time-series analysis. **Introduction to Text Analysis :** Text-preprocessing, Bag of words, TF-IDF and topics. Need and Introduction to social network analysis, Introduction to business analysis. **Model Evaluation and Selection :** Metrics for Evaluating Classifier Performance, Holdout Method and Random Sub sampling, Parameter Tuning and Optimization, Result Interpretation, Clustering and Time-series analysis using Scikit-learn, sklearn. metrics, Confusion matrix, AUC-ROC Curves, Elbow plot.

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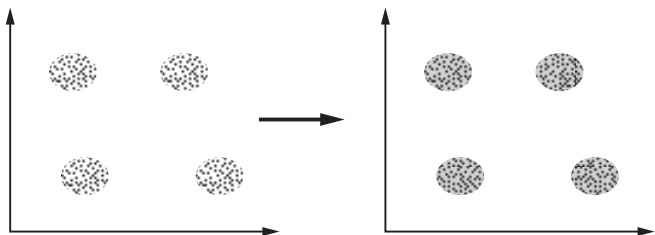
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## 5.1 Clustering Algorithms

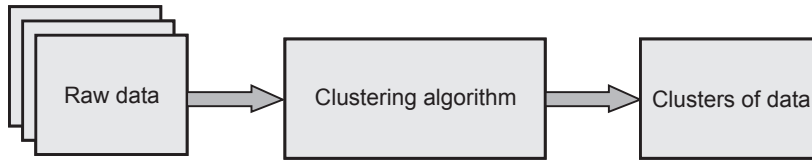
**SPPU : Aug.-18, Oct.-19, Dec.-18, 19**

### What is cluster analysis ?

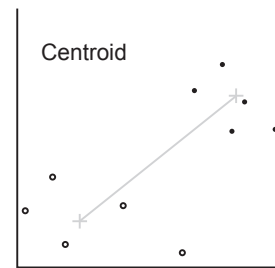
- Given a set of objects, place them in groups such that the objects in a group are similar (or related) to one another and different from (or unrelated to) the objects in other groups.
- Cluster analysis can be a powerful data - mining tool for any organisation that needs to identify discrete groups of customers, sales transactions or other types of behaviors and things. For example, insurance providers use cluster analysis to detect fraudulent claims, and banks use it for credit scoring.
- Cluster analysis uses mathematical models to discover groups of similar customers based on the smallest variations among customers within each group.
- Cluster is a group of objects that belong to the same class. In other words the similar object are grouped in one cluster and dissimilar are grouped in other cluster.
- Clustering is a process of partitioning a set of data in a set of meaningful subclasses. Every data in the subclass shares a common trait. It helps a user understand the natural grouping or structure in a data set.
- Various types of clustering methods are partitioning methods, hierarchical clustering, Fuzzy clustering, Density -based clustering and Model -based clustering.
- Cluster analysis is process of grouping a set of data - objects into clusters.
- **Desirable properties of a clustering algorithm are as follows :**
  1. Scalability (in terms of both time and space)
  2. Ability to deal with different data types
  3. Minimal requirements for domain knowledge to determine input parameters.
  4. Interpretability and usability.
- Clustering of data is a method by which large sets of data are grouped into clusters of smaller sets of similar data. Clustering can be considered the most important unsupervised learning problem.
- A cluster is therefore a collection of objects which are "similar" between them and are "dissimilar" to the objects belonging to other clusters. Fig. 5.1.1 shows cluster.

**Fig. 5.1.1 Cluster**

- In this case we easily identify the 4 clusters into which the data can be divided ; the similarity criterion is distance : Two or more objects belong to the same cluster if they are "close" according to a given distance (in this case geometrical distance). This is called **distance-based clustering**.



- Clustering means grouping of data or dividing a large data set into smaller data sets of some similarity.
- A clustering algorithm attempts to find natural groups of components or data based on some similarity. Also, the clustering algorithm finds the centroid of a group of data sets.
- To determine cluster membership, most algorithms evaluate the distance between a point and the cluster centroids. The output from a clustering algorithm is basically a statistical description of the cluster centroids with the number of components in each cluster.
- **Cluster centroid** : The centroid of a cluster is a point whose parameter values are the mean of the parameter values of all the points in the clusters. Each cluster has a well defined centroid.
- **Distance** : The distance between two points is taken as a common metric to see the similarity among the components of a population. The commonly used distance measure is the euclidean metric which defines the distance between two points  $p = (p_1, p_2, \dots)$  and  $q = (q_1, q_2, \dots)$  is given by :



$$d = \sum_{i=1}^k (p_i - q_i)^2$$

- The goal of clustering is to determine the intrinsic grouping in a set of unlabeled data. But how to decide what constitutes a good clustering ? It can be shown that there is no absolute "best" criterion which would be independent of the final aim of the clustering. Consequently, it is the user which must supply this criterion, in such a way that the result of the clustering will suit their needs.
- Clustering analysis helps construct meaningful partitioning of a large set of objects. Cluster analysis has been widely used in numerous applications, including pattern recognition, data analysis, image processing, etc.

- Clustering algorithms may be classified as listed below :
  1. Exclusive clustering
  2. Overlapping clustering
  3. Hierarchical clustering
  4. Probabilistic clustering
- A good clustering method will produce high quality clusters with high intra-class similarity and low inter-class similarity. The quality of a clustering result depends on both the similarity measure used by the method and its implementation. The quality of a clustering method is also measured by its ability to discover some or all of the hidden patterns.

### Examples of Clustering Applications

1. **Marketing** : Help marketers discover distinct groups in their customer bases and then use this knowledge to develop targeted marketing programs.
2. **Land use** : Identification of areas of similar land use in an earth observation database.
3. **Insurance** : Identifying groups of motor insurance policy holders with a high average claim cost.
4. **Urban planning** : Identifying groups of houses according to their house type, value, and geographical location.
5. **Seismology** : Observed earth quake epicenters should be clustered along continent faults.

#### **5.1.1 Typical Requirements of Clustering in Data Mining**

1. **Scalability** : Many clustering algorithms work well on small data sets.
2. **Ability to deal with different types of attributes** : Many algorithms are designed to cluster interval-based data.
3. **Discovery of clusters with arbitrary shape** : Many clustering algorithms determine clusters based on Euclidean or Manhattan distance measures.
4. **Minimal requirements for domain knowledge to determine input parameters** : Many clustering algorithms require users to input certain parameters in cluster analysis.
5. **Ability to deal with noisy data** : Most real - world databases contain outliers or missing, unknown or erroneous data. Some clustering algorithms are sensitive to such data and may lead to clusters of poor quality.
6. **Incremental clustering and insensitivity to the order of input records** : Some clustering algorithms cannot incorporate newly inserted data into existing clustering structures.

7. **High dimensionality** : A database or a data warehouse can contain several dimensions or attributes.
8. **Constraint-based clustering** : Real-world applications may need to perform clustering under various kinds of constraints.
9. **Interpretability and usability** : Users expect clustering results to be interpretable, comprehensible and usable.

### 5.1.2 Problems with Clustering

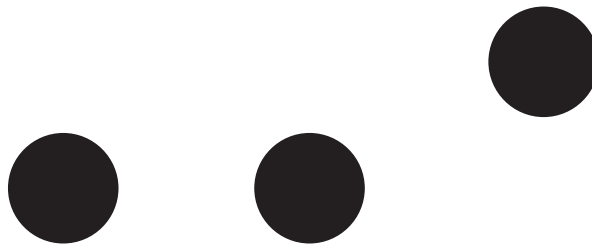
1. Current clustering techniques do not address all the requirements adequately ;
2. Dealing with large number of dimensions and large number of data items can be problematic because of time complexity ;
3. The effectiveness of the method depends on the definition of "distance" ;
4. If an obvious distance measure doesn't exist we must "define" it, which is not always easy, especially in multi-dimensional spaces ;
5. The result of the clustering algorithm can be interpreted in different ways.

### 5.1.3 Types of Clusters

- Type of clusters are as follows :
  - a) Well - separated clusters
  - b) Prototype - based clusters
  - c) Contiguity - based clusters
  - d) Density - based clusters

#### a) Well - separated clusters :

- A cluster is a set of points such that any point in a cluster is closer to every other point in the cluster than to any point not in the cluster.
- Fig. 5.1.2 shows well-separated cluster.



**Fig. 5.1.2 Well-separated cluster**

- Sometimes a threshold is used to specify that all the objects in a cluster must sufficiently close to one another. Definition of a cluster is satisfied only when the data contains natural clusters.

### b) Prototype - based cluster

- A cluster is a set of objects such that an object in a cluster is closer (more similar) to the prototype or "center" of a cluster, than to the center of any other cluster. Prototype based clusters can also be referred to as "Center-Based" Clusters.
- The center of a cluster is often a centroid, the average of all the points in the cluster, or a medoid, the most "representative" point of a cluster. Fig. 5.1.3 shows 4 center-based clusters.



**Fig. 5.1.3 4 Center-based clusters**

- If the data is numerical, the prototype of the cluster is often a centroid i.e., the average of all the points in the cluster.
- If the data has categorical attributes, the prototype of the cluster is often a medoid i.e., the most representative point of the cluster.
- Objects in the cluster are closer to the prototype of the cluster than to the prototype of any other cluster.
- K-Means and K-Medoids are the examples of Prototype-based Clustering algorithm.

### c) Contiguity - based clusters

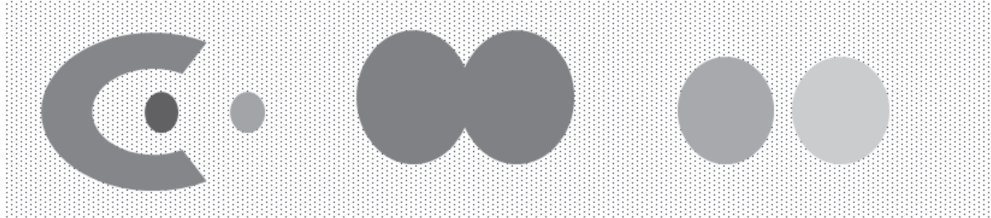
- A cluster is a set of points such that a point in a cluster is closer (or more similar) to one or more other points in the cluster than to any point not in the cluster.



**Fig. 5.1.4 8 contiguous clusters**

**d) Density - based**

- A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
- Used when the clusters are irregular or intertwined, and when noise and outliers are present.

**5.1.4 Desired Features of Cluster Analysis**

- Features are as follows :
  1. **Scalability** : Data-mining problems can be large and therefore a cluster-analysis method should be able to deal with large problems gracefully. Ideally, performance should be linear with data-size.
  2. **Only one scan of the dataset** : For large problems, data must be stored on disk, so cost of I/O disk can become significant in solving the problem.
  3. **Ability to stop and resume** : For large dataset, cluster-analysis may require huge processor-time to complete the task. In such cases, the task should be able to be stopped and then resumed when convenient.
  4. **Minimal input parameters** : The method should not expect too much guidance from the data-mining analyst.
  5. **Robustness** : Most data obtained from a variety of sources has errors. Therefore, the method should be able to deal with noise, outlier and missing values gracefully.
  6. **Ability to discover different cluster-shapes** : Clusters appear in different shapes and not all clusters are spherical. So the method should be able to discover cluster-shapes other than spherical.
  7. **Different data types** : Many problems have a mixture of data types, for e.g. numerical, categorical and even textual. Therefore, the method should be able to deal with numerical, boolean and categorical data.
  8. **Result independent of data input order** : The method should not be sensitive to data input-order.

### 5.1.5 K-Means

- K-Means clustering is heuristic method. Here each cluster is represented by the center of the cluster. "K" stands for number of clusters, it is typically a user input to the algorithm ; some criteria can be used to automatically estimate K.
- This method initially takes the number of components of the population equal to the final required number of clusters. In this step itself the final required number of clusters is chosen such that the points are mutually farthest apart.
- Next, it examines each component in the population and assigns it to one of the clusters depending on the minimum distance. The centroid's position is recalculated everytime a component is added to the cluster and this continues until all the components are grouped into the final required number of clusters.
- Given K, the K-means algorithm consists of four steps :
  1. Select initial centroids at random.
  2. Assign each object to the cluster with the nearest centroid.
  3. Compute each centroid as the mean of the objects assigned to it.
  4. Repeat previous 2 steps until no change.
- The  $x_1, \dots, x_N$  are data points or vectors of observations. Each observation (vector  $x_i$ ) will be assigned to one and only one cluster. The  $C(i)$  denotes cluster number for the observation. K-means minimizes within-cluster point scatter :

$$\begin{aligned}
 W(C) &= \frac{1}{2} \sum_{K=1}^K \sum_{C(i)=K} \sum_{C(j)=K} \|x_i - x_j\|^2 \\
 &= \sum_{K=1}^K N_K \sum_{C(i)=K} \|x_i - m_K\|^2
 \end{aligned}$$

Where,

$m_K$  is the mean vector of the  $K^{\text{th}}$  cluster.

$N_K$  is the number of observations in  $K^{\text{th}}$  cluster.

#### K-means algorithm properties

1. There are always K clusters.
2. There is always at least one item in each cluster.
3. The clusters are non-hierarchical and they do not overlap.
4. Every member of a cluster is closer to its cluster than any other cluster because closeness does not always involve the 'center' of clusters.

## The K-means algorithm process

1. The dataset is partitioned into K clusters and the data points are randomly assigned to the clusters resulting in clusters that have roughly the same number of data points.
2. For each data point,
  - a. Calculate the distance from the data point to each cluster.
  - b. If the data point is closest to its own cluster, leave it where it is.
  - c. If the data point is not closest to its own cluster, move it into the closest cluster.
3. Repeat the above step until a complete pass through all the data points results in no data point moving from one cluster to another. At this point the clusters are stable and the clustering process ends.
4. The choice of initial partition can greatly affect the final clusters that result, in terms of inter-cluster and intracluster distances and cohesion.
  - K-means algorithm is iterative in nature. It converges, however only a local minimum is obtained. It works only for numerical data. This method easy to implement.

### Advantages of K-means algorithm :

1. Efficient in computation
2. Easy to implement

### Weaknesses :

1. Applicable only when mean is defined.
2. Need to specify K, the number of clusters, in advance.
3. Trouble with noisy data and outliers.
4. Not suitable to discover clusters with non-convex shapes.

## 5.1.6 Hierarchical Clustering

- This method use distance matrix as clustering criteria. This method does not require the number of clusters K as an input, but needs a termination condition. Hierarchical clustering is a widely used data analysis tool.
- The idea is to build a binary tree of the data that successively merges similar groups of points. Visualizing this tree provides a useful summary of the data.

- Hierarchical clustering arranges items in a hierarchy with a tree like structure based on the distance or similarity between them. The graphical representation of the resulting hierarchy is a tree - structured graph called a **dendrogram**.
- Hierarchical clustering methods can be further classified as either agglomerative or divisive, depending on whether the hierarchical decomposition is formed in a bottom - up (merging) or top - down (splitting) fashion.

### Agglomerative hierarchical clustering

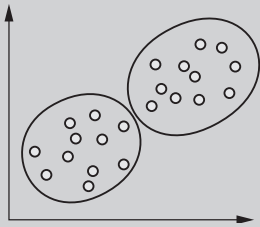
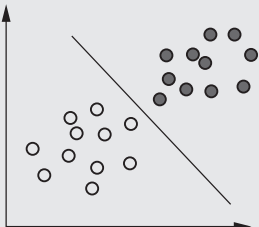
- This bottom-up strategy starts by placing each object in its own cluster and then merges these atomic clusters into larger and larger clusters, until all of the objects are in a single cluster or until certain termination conditions are satisfied. Most hierarchical clustering methods belong to this category.
- Initially, AGNES places each objects into a cluster of its own. The clusters are then merged step-by-step according to some criterion. For example, cluster  $C_1$  and  $C_2$  may be merged if an object in  $C_1$  and object in  $C_2$  form the minimum euclidean distance between any two objects from different clusters.
- In the agglomerative hierarchical approach, we start by defining each data point to be a cluster and combine existing clusters at each step. Here are four different methods for doing this :
  1. **Single linkage** : Smallest pairwise distance between elements from each cluster.
  2. **Complete linkage** : Largest distance between elements from each cluster.
  3. **Average linkage** : The average distance between elements from each cluster.
  4. **Centroid linkage** : Distance between cluster means.

### Divisive hierarchical clustering

- This top-down strategy does the reverse of agglomerative hierarchical clustering by starting with all objects in one cluster. It subdivides the clusters into smaller and smaller pieces, until each object form a cluster on its own or until it satisfies certain termination conditions, such as a desired number of cluster or the diameter of each cluster is within a certain threshold.

| Agglomerative                             | Divisive hierarchical clustering         |
|-------------------------------------------|------------------------------------------|
| Initially each item in its own cluster.   | Initially all items in one cluster.      |
| Iteratively clusters are merged together. | Large clusters are successively divided. |
| Bottom up.                                | Top down.                                |

### 5.1.7 Difference between Clustering vs Classification

| Sr. No. | Clustering                                                                                                                           | Classification                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1.      | This function maps the data into one of several clusters which is the grouping of data items based on the similarities between them. | This model function classifies the data into one of several predefined categorical classes.             |
| 2.      | Involved in unsupervised learning                                                                                                    | Involved in supervised learning                                                                         |
| 3.      | Training sample is not provided.                                                                                                     | Training sample is provided.                                                                            |
| 4.      | The number of cluster is not known before clustering, These are identified after the completion of clustering.                       | The number of classes is known before classification as there is predefined output based on input data. |
| 5.      | Data is not labeled.                                                                                                                 | Labeled data points.                                                                                    |
| 6.      | Asks how can I group this set of items ?                                                                                             | Asks what class does this item belong to ?                                                              |
| 7.      | Unknown number of classes                                                                                                            | Known number of classes                                                                                 |
| 8.      | Used to understand data                                                                                                              | Used to classify future observations                                                                    |
| 9.      |                                                    |                      |

#### Review Questions

1. What is clustering ? Explain *k* - means clustering algorithm with the use cases.

**SPPU : Aug.-18 (In Sem), Marks 6**

2. Explain *k*-means clustering algorithm. What are its drawbacks ?

**SPPU : Dec.-18 (End Sem), Marks 7**

3. Explain *k*-means algorithm.

**SPPU : Oct.-19 (In Sem), Marks 5**

4. How *k* - means algorithm works ?

**SPPU : Dec.-19 (End Sem), Marks 5**

## 5.2 Time-Series Analysis

- Time series is a sequence of data points in chronological sequence, most often gathered in regular intervals. Time series analysis can be applied to any variable that changes over time and generally speaking, usually data points that are closer together are more similar than those further apart.

- Time series analysis is the way of studying the characteristics of the response variable with respect to time, as the independent variable. To estimate the target variable in the name of predicting or forecasting, use the time variable as the point of reference.
- Components of time series analysis are trend, seasonality, cyclical and irregularity.
  - a) **Trend** : In which there is no fixed interval and any divergence within the given dataset is a continuous timeline. The trend would be negative or positive or null trend.
  - b) **Seasonality** : In which regular or fixed interval shifts within the dataset in a continuous timeline. Would be bell curve or saw tooth.
  - c) **Cyclical** : In which there is no fixed interval, uncertainty in movement and its pattern.
  - d) **Irregularity** : Unexpected situations/events/scenarios and spikes in a short time span.
- Identifying seasonality in time series data is important for the development of a useful time series model.
- There are many tools that are useful for detecting seasonality in time series data :
  1. Background theory and knowledge of the data can provide insight into the presence and frequency of seasonality.
  2. Time series plots such as the seasonal subseries plot, the autocorrelation plot, or a spectral plot can help identify obvious seasonal trends in data.
  3. Statistical analysis and tests, such as the autocorrelation function, periodograms, or power spectrums can be used to identify the presence of seasonality.
- Time series data are data points collected over a period of time as a sequence of time gap. Time series data analysis means analyzing the available data to find out the pattern or trend in the data to predict some future values which will, in turn, help more effective and optimize business decisions.
- A forecast "error" is the difference between an observed value and its forecast.
- Time series analysis can be classified as :
  1. Parametric and Non-parametric
  2. Linear and Non-linear and
  3. Univariate and Multivariate
- Techniques used for time series analysis :
  1. ARIMA models
  2. Box-Jenkins multivariate models
  3. Holt winters exponential smoothing (single, double and triple)

### 5.2.1 ARIMA

- ARIMA stands for AutoRegressive Integrated Moving Average. It is a forecasting technique that projects the future values of a series based entirely on its own inertia.
- Its main application is in the area of short term forecasting requiring at least 40 historical data points. It works best when data exhibits a stable or consistent pattern over time with a minimum amount of outliers.
- Parameters of the ARIMA model :
  1. **p (lag order)** : Number of lag observations included in the model.
  2. **d (degree of differencing)** : Number of times that the raw observations are differenced.
  3. **q (order of moving average)** : Size of the moving average window.

#### Assumptions of ARIMA model :

1. **Data should be stationary** : By stationary it means that the properties of the series doesn't depend on the time when it is captured. A white noise series and series with cyclic behaviour can also be considered as stationary series.
  2. **Data should be univariate** : ARIMA works on a single variable. Auto-regression is all about regression with the past values.
- **AR (Autoregression)** : A model that uses the dependent relationship between an observation and some number of lagged observations. 'p' is used as a notation to define this component, called the lag order.
  - **I (Integrated)** : The use of differencing of raw observations (e.g. subtracting an observation from an observation at the previous time step) in order to make the time series stationary. 'd' is used as a notation to define this component, called the degree of differencing.
  - **MA (Moving Average)** : A model that uses the dependency between an observation and a residual error from a moving average model applied to lag observations. 'q' is used as a notation to define this component, called the order of moving average.

#### ARIMA forecasting equation :

- Let Y denote the original series and y denote the differenced (stationaries) series.

No difference ( $d = 0$ ) :  $y_t = Y_t$

First difference ( $d = 1$ ) :  $y_t = Y_t - Y_{t-1}$

Second difference ( $d = 2$ ) :

$$\begin{aligned} y_t &= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) \\ &= Y_t - 2Y_{t-1} + Y_{t-2} \end{aligned}$$

- Basic model for ARIMA is as follows :

1. **ARIMA (1, 0, 0)** : First-order autoregressive model-if the series is stationary and auto correlated, perhaps it can be predicted as a multiple of its own previous value, plus a constant.

$$y_t = \beta_0 + \beta_1 Y_{t-1}$$

2. **ARIMA (2, 0, 0)** : The series can be said to be dependent on its previous two values.

$$y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2}$$

3. **ARIMA (0, 1, 0)** : Random walk : If the series  $Y$  is not stationary, the simplest possible model for it is a random walk model, which can be considered as a limiting case of an AR(1) model in which the auto-regressive coefficient is equal to 1, i.e. series with infinitely slow mean reversion.

$$y_t - y_{t-1} = \mu$$

4. **ARIMA (1, 1, 0)** : Differenced first-order autoregressive model : If the errors of a random walk model are auto correlated, perhaps the problem can be fixed by adding one lag of the dependent variable to the prediction equation i.e., by regressing the first difference of  $Y$  on itself lagged by one period. This would yield the following prediction equation.

$$y_t - y_{t-1} = \mu + \beta_1 (Y_{t-1} - Y_{t-2})$$

5. **ARIMA (0,1,1)** : Without constant moving average model and can expressed as -

$$y_t - y_{t-1} = \phi_1 (\alpha_{t-1} - \alpha_{t-2})$$

### 5.2.2 STL Approach

- STL is an acronym for "Seasonal and Trend decomposition using Loess.
- Seasonal trend decomposition using loess (STL) is a method for decomposing a time series into seasonal, trend and remainder components using local regression techniques.
- The goal of STL is to decompose a time series,  $y_i$ ,  $i = 1, \dots, n$  into seasonal, trend and remainder components  $s_i, t_i, r_i$  respectively.

$$y_i = s_i + t_i + r_i$$

- STL works by iterating through smoothing of the seasonal and trend components. The seasonal component is defined based on the periodicity of the time series and how many observations there are per period.
- All smoothing operations done in STL. STL is a procedure for regular time series, so the design points of the smoothing operation are equally-spaced, and without loss of generality can thought of as 1, ..., n.
- For time series smoothing, q is always an odd integer. Each smoothing operation in STL requires smoothing parameters to be specified, but many of them have nice defaults.

### **5.3 Introduction to Text Analysis**

- Text Analysis (TA) aims to extract machine-readable information from unstructured text in order to enable data-driven approaches towards managing content. The purpose of Text Analysis is to create structured data out of free text content.
- Text analysis, also known as text mining, is the process of automatically classifying and extracting meaningful information from unstructured text.
- Text mining is understood as a process of automatically extracting meaningful, useful, previously unknown and ultimately comprehensible information from textual document repositories.
- Text mining can be visualized as consisting of two phases : Text refining that transforms free-form text documents into a chosen intermediate form, and knowledge distillation that deduces patterns or knowledge from the intermediate form.
- A process of text mining involves a series of activities to be performed to mine the information. These activities are :

**1. Text Pre-processing :** It involves a series of steps as shown in below :

- i. Text Clean-up :** Text Clean-up means removing any unnecessary or unwanted information. Such as remove ads from web pages, normalize text converted from binary formats.
- ii. Tokenization :** Tokenizing is simply achieved by splitting the text into white spaces.
- iii. Part of Speech Tagging :** Part-of-Speech (POS) tagging means word class assignment to each token. Its input is given by the tokenized text. Taggers have to cope with unknown words (OOV problem) and ambiguous word-tag mappings.

**2. Text Transformation (Attribute Generation) :** A text document is represented by the words it contains and their occurrences. Two main approaches to document representation are : Bag of words and Vector Space.

3. **Feature Selection (Attribute Selection)** : Feature selection also is known as variable selection. It is the process of selecting a subset of important features for use in model creation. Redundant features are the one which provides no extra information. Irrelevant features provide no useful or relevant information in any context.
  4. **Data Mining** : At this point, the text mining process merges with the traditional process. Classic data mining techniques are used in the structured database. Also, it resulted from the previous stages.
  5. **Evaluate** : Evaluate the result, after evaluation, the result can be discarded.
- The five fundamental steps involved in text mining are :
    1. Gathering unstructured data from multiple data sources like plain text, web pages, pdf files, emails, and blogs, to name a few.
    2. Detect and remove anomalies from data by conducting pre-processing and cleansing operations. Data cleansing allows user to extract and retain the valuable information hidden within the data and to help identify the roots of specific words.
    3. Convert all the relevant information extracted from unstructured data into structured formats.
    4. Analyse the patterns within the data via Management Information System (MIS).
    5. Store all the valuable information into a secure database to drive trend analysis and enhance the decision-making process of the organization.

### **5.3.1 Use of a Text Mining Tool**

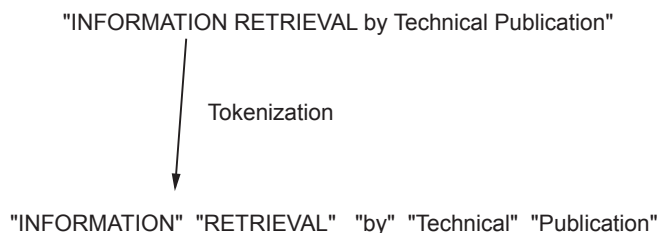
1. **Text analytics** : Involves extracting useful information and patterns from text. Most tools provide this feature.
2. **Text processing** : Involves transforming and manipulating unstructured text so that analysis methods can be applied to it.
3. **Classification/Categorization** : Many tools are used for classification and categorization of text/documents.
4. **Sentiment analysis** : Is used to identify subjective information from text. Many tools provide for sentiment analysis also called as opinion mining.
5. **Knowledge discovery** : Deals with identification of useful information from huge amount of text. Most tools provide for knowledge discovery and information retrieval features.

### 5.3.2 Text Pre - processing

- Text pre - processing is required to transform the text into an understandable format so that machine learning algorithms can be applied to it.
- As we know machine learning needs data in the numeric form. We basically used encoding technique to encode text into numeric vector. But before encoding we first need to clean the text data and this process to prepare or clean text data before encoding is called text pre-processing.
- The various text pre-processing steps are : Tokenization, Lower casing, Stop words removal, Stemming and Lemmatization.

#### 5.3.2.1 Tokenization

- Tokenization is breaking the raw text into small chunks. Tokenization breaks the raw text into words, sentences called tokens. The simplest form of analysis is to reduce different word forms into tokens.
- Tokenization is a way of separating a piece of text into smaller units called tokens. Here, tokens can be either words, characters, or subwords.
- Tokenization can be broadly classified into 3 types : Word, character, and subword (n-gram characters) tokenization.
- These tokens help in understanding the context or developing the model for the NLP. The tokenization helps in interpreting the meaning of the text by analyzing the sequence of the words.



- Tokenization can be done to either separate words or sentences. A token is an instance of a sequence of characters in some particular document that are grouped together as a useful semantic unit for processing.
- Tokens are the building blocks of natural language, the most common way of processing the raw text happens at the token level.
- The major question of the tokenization phase is what are the correct tokens to use ? You chop on whitespace and throw away punctuation characters.
- Types of tokenization are white space, dictionary based, rule based, penn tree, spacy, subword etc.

### 5.3.2.2 Stemming

- Stemming is the process for reducing inflected words to their stem, base or root form, generally a written word form. Stemming allows a query term such as "orienteering" to match an occurrence of "orienteers", or "runs" to match "running".
- For example, "orienteering" and "orienteers" might reduce to the root form "orienteer"; "runs" and "running" might reduce to "run".
- In an IR system a stemmer may be applied at both indexing time and query time. During indexing each token is passed through the stemmer and the resulting root form is indexed.
- At query time, the query terms are passed through the same stemmer and matched against the index terms. Thus the query term "runs" would match an occurrence of "running" by way of their common root form "run".
- Stemming is the particular case of tokenization which reduces inflected forms to a single base form or stem. Stemming algorithms are basic string - handling algorithms, which depend on rules which identify affixes that can be stripped.
- The goal of both stemming and lemmatization is to reduce inflectional forms and sometimes derivationally related forms of a word to a common base form. For instance :  
    am, are, is  $\Rightarrow$  be  
    car, cars, car's, cars'  $\Rightarrow$  car
- The result of this mapping of text will be something like : The boy's cars are different colors the boy car be differ color.

### 5.3.2.3 Stop Words

- The removal of high frequency words, 'stop' words or 'fluff' words is one way of implementing Luhn's upper cut-off. This is normally done by comparing the input text with a 'stop list' of words which are to be removed.
- Sometimes, some extremely common words that would appear to be of little value in helping select documents matching a user need are excluded from stop words the vocabulary entirely. These words are called stop words.
- Function words are words that have no well - defined meanings in and of themselves. Function words are usually among the most frequently occurring words in any language. At query time these stopwords are stripped from the query, and retrieval takes place on the basis of the remaining terms alone.
- Examples of a few stop words in English are "the", "a", "an", "so", "what".
- Removal of stop words definitely reduces the dataset size and thus reduces the training time due to the fewer number of tokens involved in the training.

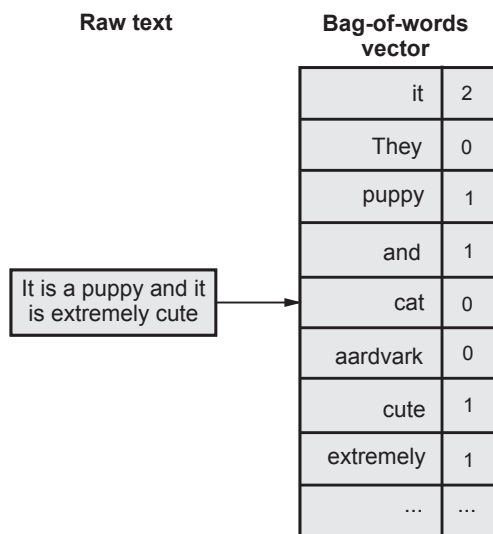
- The phrase query "Capital of the India," which contains two stop words, is more precise than Capital AND "India".
- Stop word list is as follows :  
*a an and are as at be by for from has he in is it its of on that the to was were will with*
- With stopwords present in the index, the IR system can make a decision on a query - by - query basis. Ranking methods in a modern commercial search engine will incorporate many ranking features, including features based on term frequency and proximity.
- For features that do not consider proximity between query terms, stopwords may be eliminated. For features that do consider proximity between query terms, particularly to match their occurrence in phrases, it may be appropriate to retain stopwords.
- Good compression techniques means the space for including stopwords in a system is very small.
- Good query optimization techniques mean you pay little at query time for including stopwords.

#### **5.3.2.4** Lemmatization

- Many complex or technical concepts and many organization and product names are multiword compounds or phrases. An information retrieval system uses phrases to index, retrieve, organize and describe documents.
- Phrases are identified that predict the presence of other phrases in documents. Documents are indexed according to their included phrases.
- Most recent search engines support a double quotes syntax ("technical publications") for phrase queries, which has proven to be very easily understood and successfully used by users.
- One approach to handling phrases is to consider every pair of consecutive terms in a document as a phrase. For example the text *Friends, Romans, Countrymen* would generate the biwords :  
*friends romans*  
*romans countrymen*
- Each of these biwords is now a dictionary term. Two - word phrase query - processing is now immediate.
- Biword indexes are not the standard solution but can be part of a compound strategy.

### 5.3.3 Bag of Words

- Bag of words model helps convert the text into numerical representation such that the same can be used to train models using machine learning algorithms.
- The bag-of-words model is commonly used in methods of document classification where the occurrence of each word is used as a feature for training a classifier.
- Here are the key steps of fitting a bag-of-words model :
  1. Create a vocabulary index of words or tokens from the entire set of documents. The vocabulary indices can be created in alphabetical order.
  2. Construct the numerical feature vector for each document that represents how frequent each word appears in different documents. The feature vector representing each will be sparse in nature as the words in each document will represent only a small subset of words out of all words present in entire set of documents.
- Fig. 5.3.1 shows turning raw text into a bag of words representation.



**Fig. 5.3.1 Turning raw text into a bag of words representation**

- Bag of words simply refers to a matrix in which the rows are documents and the columns are words. The values matching a document with a word in the matrix, could be a count of word occurrences within the document or use tf-idf.
- Classifiers are used to train the bag of words and a special kind of algorithm used to break words down into categories.
- Traditionally, text documents are represented in NLP as a bag-of-words. This means that each document is represented as a fixed-length vector with length equal to the vocabulary size.

- Each dimension of this vector corresponds to the count or occurrence of a word in a document. Being able to reduce variable-length documents to fixed-length vectors makes them more amenable for use with a large variety of Machine Learning (ML) models and tasks.

### 5.3.4 TF-IDF Weighting

- Term Frequency (TF) : Frequency of occurrence of query keyword in document.
- Inverse Document Frequency (IDF) : How many documents the query keyword occurs in.
- Inverse Document Frequency (IDF) is a popular measure of a word's importance. It's defined as the logarithm of the ratio of number of documents in a collection to the number of documents containing the given word. This means rare words have high IDF and common words have low IDF.
- Term frequency is a measure of the importance of terms  $i$  in document  $j$ .
- Inverse document frequency is a measure of the *general* importance of the term.
- High term frequency for "apple" means that apple is an important word in a specific document. But high document frequency (low inverse document frequency) for "apple", given a particular set of documents, means that apple is not all that important overall, since it is in all of the documents.
- The weight increases as the number of documents in which the term appears decreases. High value indicates that the word occurs more often in this document than average.
- The term frequency  $tf_{t,d}$  of term  $t$  in document  $d$  is defined as the number of times that  $t$  occurs in  $d$ .
- A document with  $tf = 10$  occurrences of the term is more relevant than a document with  $tf = 1$  occurrence of the term. But not 10 times more relevant. Relevance does not increase proportionally with term frequency.
- The document frequency is the number of documents in the collection that the term occurs in. We define the idf weight of term  $t$  as follows :

$$\text{idf weight (idf}_t\text{)} = \log 10 \frac{N}{df_t}$$

here  $N$  is the number of documents in the collection

- The tf-idf weight of a term is the product of its tf weight and its idf weight

$$W_{t,d} = (1 + \log tf_{t,d}) \cdot \log \frac{N}{df_t}$$

**Stop lists and Stemming :**

- **Stoplists** : This is a list of words that we should ignore when processing documents, since they give no useful information about content.
- **Stemming** : This is the process of treating a set of words like "fights, fighting, fighter, ..." as all instances of the same term - in this case the stem is "fight".

**5.4 Need and Introduction to Social Network Analysis**

- Social Network Analysis [SNA] is the mapping and measuring of relationships and flows between people, groups, organizations, computers, URLs, and other connected information/knowledge entities. The term "social network" has been introduced by Barnes in 1954.
- SNA is the study of social relations among a set of actors. The methods of data collection in network analysis are aimed at collecting relational data in a reliable manner. Data collection is typically carried out using standard questionnaires and observation techniques that aim to ensure the correctness and completeness of network data.
- Social network analysis is based on an assumption of the importance of relationships among interacting units. The social network perspective encompasses theories, models, and applications that are expressed in terms of relational concepts or processes.
- The nodes in the network are the people and groups while the links show relationships or flows between the nodes. SNA provides both a visual and a mathematical analysis of human relationships.
- The advantage of social network analysis is that, unlike many other methods, it focuses on interaction. Network analysis allows us to examine how the configuration of networks influences how individuals and groups, organizations, or systems function.
- **Features of social network analysis** : Structural intuition, systematic relational data, graphic representation and mathematical or computational models.

**Fundamental concepts in network analysis :**

Following terminology is used in social network analysis.

- |             |                   |             |          |
|-------------|-------------------|-------------|----------|
| 1. Actor    | 2. Relational tie | 3. Dyad     | 4. Triad |
| 5. Subgroup | 6. Group          | 7. Relation |          |

- a) Actor** : Actor is discrete individual, corporate, or collective social units. Examples : people in a group, departments within in a corporation, public service agency in a city, nation - states in the world system.

- b) Relational tie :** Actors are linked to another by social ties. A tie establishes a linkage between a pair of actors.
- c) Dyad :** It is a tie between two actors and consists of a pair of actors and the tie(s) between them.
- d) Triad :** Triples of actors and associated ties. A subset of three actors and the tie(s) among them.
- e) Subgroup** of actors is defined as any subset of actors, and all ties among them.
- f) Group :** Group is the collection of all actors on which ties are to be measured.
- g) Relation :** It is the collection of ties of a specific kind among members of a group.  
Example : The set of friendship among pairs of children in a classroom.
- Network can be categorized by the nature of the sets of actors and the properties of the ties among them. The number of modes in a network refers to the number of distinct kinds of social entities in the network.
- One-mode networks are a single set of actors. Two-mode networks are focus on two sets of actors, or one set of actors and one set of events.

### Principles of social network analysis

1. Actors and their actions are viewed as interdependent rather than independent, autonomous units.
2. Relational ties (linkages) between actors are channels for transfer or "flow" of resources (either material or nonmaterial).
3. Network models focusing on individuals view the network structure environment as providing opportunities for or constraints on individual action.
4. Network models conceptualize structure (social, economic, political, and so forth) as lasting patterns of relations among actors.

#### **5.4.1** Development of Social Network Analysis

- A social network is a group of collaborating, and/or competing individuals or entities that are related to each other. It may be presented as a graph, or a multi-graph; each participant in the collaboration or competition is called an actor and depicted as a node in the graph theory.
- Valued relations between actors are depicted as links, or ties, either directed or undirected, between the corresponding nodes.
- Actors can be persons, organizations, or groups - any set of related entities. As such, SNA may be used on different levels, ranging from individuals, web pages, families, small groups, to large organizations, parties, and even to nations.

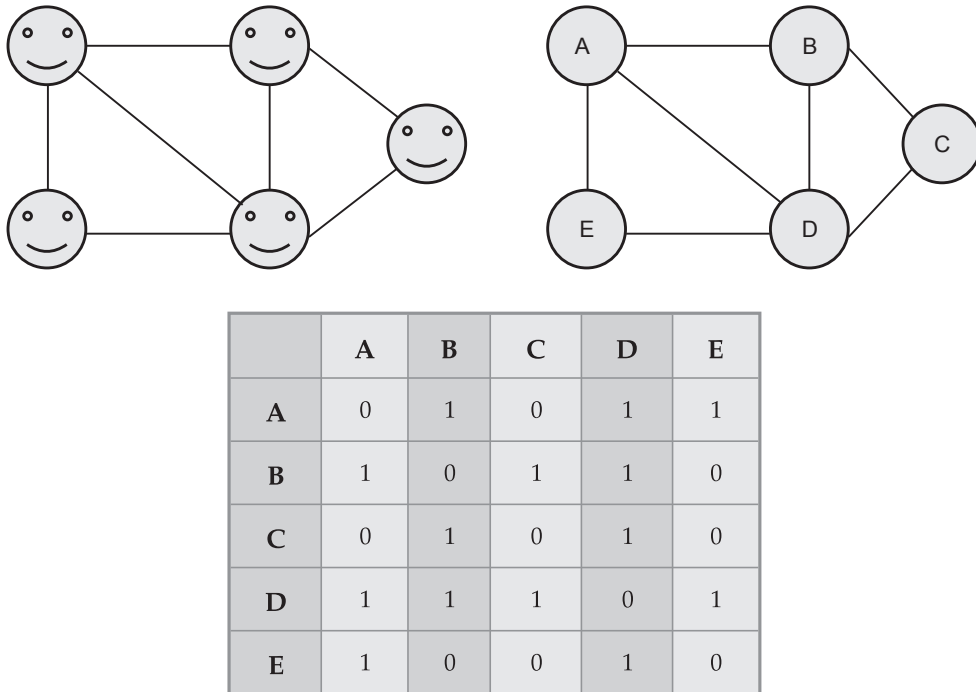
- In general, a social network consists of actors (e.g., persons, organizations) and some form of relation among them. The network structure is usually modeled as a graph, in which vertices represent actors, and edges represent ties, i.e., the existence of a relation between two actors.
- The vocabulary, models and methods of network analysis also expand continuously through applications that require to handle ever more complex data sets.
- An example of this process are the advances in dealing with longitudinal data. New probabilistic models are capable of modelling the evolution of social networks and answering questions regarding the dynamics of communities. Formalizing an increasing set of concepts in terms of networks also contributes to both developing and testing theories in more theoretical branches of sociology.
- The purpose of social network analysis is to identify important actors, crucial links, roles, dense groups, and so on, in order to answer substantive questions about structure.
- Analysis methods are divided into four main categories according to the level or subject of interest : vertex, dyad, group, and network level.
- Available analysis methods include actor-level centrality indices, e.g. closeness, betweenness, and pagerank, cohesive subgroups like cliques, k-cliques, and k-clans, centrality and connectedness.
- These levels break further down into measures of the same objective, e.g., connectedness or cohesiveness. Analysis methods are accessible using the analysis tab in the control area.

### **Key concepts and measures in network analysis**

- Social network analysis has developed a set of concepts and methods specific to the analysis of social networks.
- Several analytic tendencies distinguish social network analysis :
  1. There is no assumption that groups are the building blocks of society : The approach is open to studying less - bounded social systems, from nonlocal communities to links among websites.
  2. Rather than treating individuals (persons, organizations, states) as discrete units of analysis, it focuses on how the structure of ties affects individuals and their relationships.
  3. In contrast to analyses that assume that socialization into norms determines behavior, network analysis looks to see the extent to which the structure and composition of ties affect norms.

### 5.4.2 Global Structure of Networks

- Social network can be represented as a graph  $G = (V, E)$   
 where  $V$  = The finite set of vertices  
 $E$  = Finite set of edges such
- The most network analysis methods work on an abstract, graph based representation of real world networks. It is shown in Fig. 5.4.1.



**Fig. 5.4.1 Graph based representation of real world networks**

- When representing a network as a graph, all of the connections are pair-wise and hence represented by ties known as edges.
- Networks can be described using a mixture of local, global, and intermediate-scale perspectives. Accordingly, one of the key uses of network theory is the identification of summary statistics for large networks in order to develop a framework for analyzing and comparing complex structures.
- SNA can produce maps like the one featured below, and provide statistical measures of relationships between actors. In SNA maps, the nodes represent the different actors in the network, and the lines represent the relationships between the various actors.

- The size of the node often represents the relative importance of that actor in the network, and the thickness of the connecting line denotes the strength of the relationship.
- Clustering for a single vertex can be measured by the actual number of the edges between the neighbors of a vertex divided by the possible number of edges between the neighbors.
- When taken the average over all vertices, we get to the measure known as clustering coefficient. The clustering coefficient of a tree is zero, which is easy to see if we consider that there are no triangles of edges (triads) in the graph. In a tree, it would never be the case that our friends are friends with each other.
- The coordination degree measures the ability of the vertices in a graph to interchange information. There are several ways in which we can model this magnitude. One of the easiest is to consider the coordination degree to be exponentially related with the distance between the vertices.
- To define the total co-ordination degree of a vertex "i" in a graph as the sum of all the coordination degrees between that particular vertex and the rest :

$$\Gamma_i = \sum_{j=1}^N \gamma_{ij}$$

Where N is the order of the graph.

- Graph density (D) is defined as the total number of observed lines in a graph divided by the total number of possible lines in the same graph. Density ranges from 0 to 1.

$$\text{Density (D)} = \frac{\text{Number of lines (L)}}{(\text{Number of points (Number of points - 1))} / 2} = \frac{2L}{g(g-1)}$$

### 5.4.3 Random Graphs with Arbitrary Degree Distributions

- A random graph is simple to define. One takes some number N of nodes or "vertices" and places connections or "edges" between them, such that each pair of vertices i, j has a connecting edge with independent probability p.
- Random graph can be generated by taking a set of vertices with no edges connecting them. Subsequently, edges are added by picking pairs of nodes with equal probability.

- Consider a vertex in a random graph. It is connected with equal probability  $p$  with each of the  $N - 1$  other vertices in the graph, and hence the probability  $p_k$  that it has degree exactly  $k$  is given by the binomial distribution :

$$p_k = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

- A large random graph has a Poisson degree distribution. This degree distribution makes the random graph a poor approximation to the real-world networks.

#### **5.4.4 Macro-Structure of Social Networks**

- Network visualizations based on topographic or physical principles can be helpful in understanding the group structure of social networks and pinpoint hubs that naturally tend to gravitate toward the center of the visualization.
  - Clustering a graph into subgroups allows us to visualize the connectivity at a group level.
  - Core-Periphery structure is one where nodes can be divided in two distinct subgroups : Nodes in the core are densely connected with each other and the nodes on the periphery, while peripheral nodes are not connected with each other, only nodes in the core.
  - By computing a network's core-periphery structure, one attempts to determine which nodes are part of a densely connected core and which are part of a sparsely connected periphery.
  - Core nodes should also be reasonably well-connected to peripheral nodes, but the latter are not well-connected to a core or to each other.
  - Node belongs to a core if and only if it is well-connected both to other core nodes and to peripheral nodes. A core structure in a network is thus not merely densely connected but also tends to be 'central' to the network.
  - From network theory has it defined the dual relationships between nodes in the network, so that if an agent has a feature no other, for example, if it is good then it is not bad, is a bipartition graph in which each element of a subset is additional to another concept indeed implies that binding of the  $n$  subgroups partitions make the whole graph. So CPS, involves dividing the nodes of the network into two groups.
  - Fig. 5.4.2 shows core-periphery structure that would be perfect without the edge between nodes.
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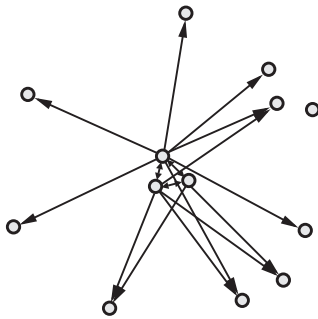


Fig. 5.4.2 (a) : Core - periphery structure

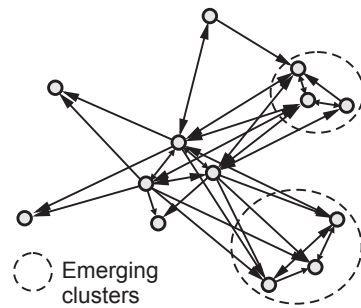


Fig. 5.4.2 (b) Cluster structure

### 1. Affiliation network :

- **Affiliation networks** contain information about the relationships between two sets of nodes : A set of subjects and a set of affiliations. An affiliation network can be formally represented as a bipartite graph, also known as a two-mode network.
- Affiliation networks are **two mode networks** that allow one to study the dual perspectives of the actors and the events. They look at collections or subsets of actors or subsets rather than ties between pairs of actors. Connections among members of one of the modes as based on linkages established through the second mode.
- An affiliation network is a network in which actors are joined together by common membership of groups or clubs of some kind.
- A distinctive feature of affiliation networks is **duality** i.e. events can be described as collections of individuals affiliated with them and actors can be described as collections of events with which they are affiliated.
- Based on two-mode matrix data, affiliation networks consist of sets of relations connecting actors and events, rather than direct ties between pairs of actors as in one-mode data. Familiar affiliation networks include persons belonging to associations, social movement activists participating in protest events, firms creating strategic alliances, and nations signing treaties.
- The representation of two-mode data should facilitate the visualization of three kinds of patterning :
  - a. The actor-event structure
  - b. The actor-actor structure
  - c. The event-event structure

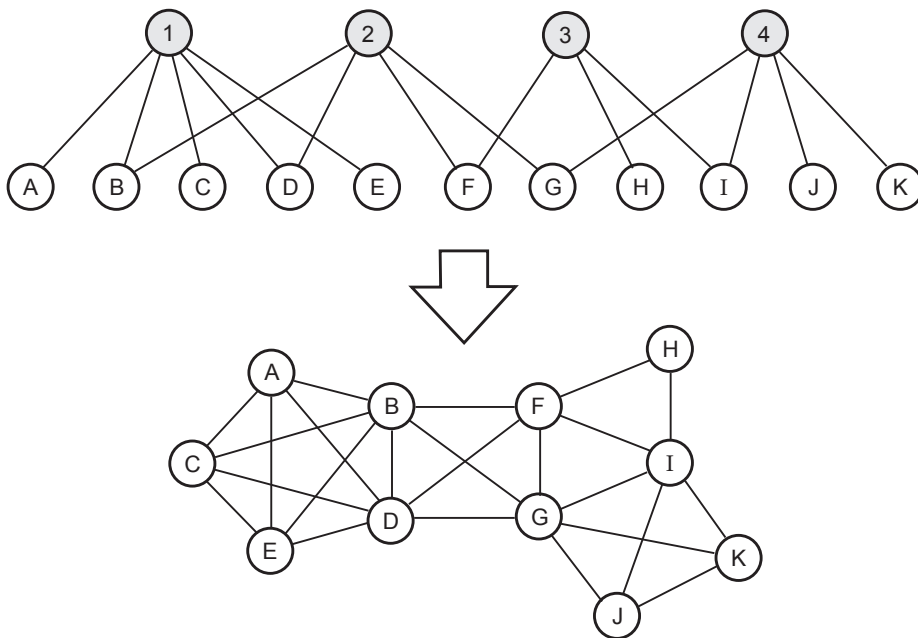
- Many ways to represent affiliation networks :
  1. Affiliation network matrix
  2. Bipartite graph
  3. Hypergraph
  4. Simplicial complex

### Benefits of affiliations network

1. Affiliations of actors with events provide a direct linkage between actors through memberships in events, or between events through common memberships.
2. Affiliations provide conditions that facilitate the formation of pairwise ties between actors.
3. Affiliations enable us to model the relationships between actors and events as a whole system.

### 2. Bipartite graph :

- Nodes are partitions into two subsets and all lines are between pairs of nodes belonging to different subsets. Fig 5.4.3 shows bipartite network. As there are  $g$  actors and  $h$  events, there are  $g + h$  nodes.



**Fig. 5.4.3 Bipartite graph**

- The lines on the graph represent the relation "is affiliated with" from the perspective of the actor and the relation "has as a member" from the perspective of the event.
- No two actors are adjacent and no two events are adjacent. If pairs of actors are reachable, it is only via paths containing one or more events. Similarly, if pairs of events are reachable, it is only via paths containing one or more actors.

### Advantages

1. They highlight the connectivity in the network, as well as the indirect chains of connection.
2. Data is not lost and we always know which individuals attended which events.

### Disadvantage

1. They can be unwieldy when used to depict larger affiliation networks.

## 5.4.5 Application of Social Network Analysis

- Social Network Analysis (SNA) is an important and valuable tool for knowledge extraction from massive and un-structured data. Social network provides a powerful abstraction of the structure and dynamics of diverse kinds of inter-personal connection and interaction.
- Facebook is a social networking service and website that connects people with other people, and share data between people. A user can create a personal profile, add other users as friends, exchange data, create and join common interest communities.
- Twitter is a social net-working and microblogging service. The users of Twitter can exchange text-based posts called tweets. A tweet is a maximum 140 characters long but can be augmented by pictures or audio recording. The main concept of Twitter was to build a social network formed by friends and followers. Friends are people who you follow, followers are those who follow you.
- The role of social networks in labor markets deserves attention for at least two reasons : First, because of the central role networks play in disseminating information about job openings they place a critical role in determining whether labor markets function efficiently; and second, because network structure ends up having implications for things like human capital investment as well as inequality.
- Social Network Analysis (SNA) primarily focuses on applying analytic techniques to the relationships between individuals and groups, and investigating how those relationships can be used to infer additional information about the individuals and groups.

- SNA is used in a variety of domains. For example, business consultants use SNA to identify the effective relationships between workers that enable work to get done; these relationships often differ from connections seen in an organizational chart.

## **5.5 Introduction to Business Analysis**

- Business Analysis is a discipline and practice of defining business needs and recommending solutions to business problems. Business analysis deals with the current state of each company, desired future state, stakeholders' needs, processes, software and more.
- A business intelligence system provides decision makers with information and knowledge extracted from data, through the application of mathematical models and algorithms.
- The adoption of a business intelligence system tends to promote a scientific and rational approach to the management of enterprises and complex organizations.
- Obstacle to business intelligence in an organization are as follows :
  - 1. Lack of BI strategy :** Organizations should proactively define the problems they trying to solve. Only then they will be able to identify the right Business Intelligence solution that will suit their requirements.
  - 2. Business intelligence :**
    - When You Don't Know How to Code. Now a days, executives find it difficult to access the right data at right time. And even if they do find what they're looking for, data formats are typically so complex and unstructured it's hard to find out meaningful and relevant data.
    - Now unless they are using Excel extensively, they probably would not get much satisfaction (or value) from their BI system.
    - A good practice would be to replace Excel Sheets with intuitive dashboards to make data more engaging, meaningful and eventually very powerful.
  - 3. Lack of training and execution :**
    - Many a times, companies might have well-articulated requirements, a sound BI strategy, and a good tool solution, but lack technical skills like designing, building, maintaining, and supporting BI applications.
    - This results in BI applications to run slowly, break frequently, deliver uncertain results and eventually leading to rising cost of using the BI solution. The causes of lack of execution often are multiple and varied, as are its remedies.

**4. Lack of BI impact (Low utilization) :**

- Management might always wonder why there is no change in business results attributable to BI and might feel that business value of BI investments not captured. This indicates that the organization is not utilizing the BI solution at par with global standards and best practices.

**5. Business intelligence with unstructured data :**

- Most of the times data is unstructured for BI to analyze. This lead to a problem when users need to perform simple BI processes.
- Businesses may invest in big data analytics but cannot complete the tasks in time. They may result to people spending hours on cleaning and structuring the data first and then using the BI solution.

**6. Installation and deployment :**

- A painful BI solution installation and deployment would be difficult to maintain. Even an unplanned and rushed deployment would be unsuccessful so often.
- Doing this may leave users void with time to understand the system and develop the skills using the solution effectively.

**5.6 Model Evaluation and Selection****SPPU : Dec.-18**

- A binary classification rule is a method that assigns a class to an object, on the basis of its description.
- The performance of a binary classifier can be assessed by tabulating its predictions on a test set with known labels in a contingency table or confusion matrix, with actual classes in rows and predicted classes in columns.
- Measures of performance need to satisfy several criteria :
  1. They must coherently capture the aspect of performance of interest ;
  2. They must be intuitive enough to become widely used, so that the same measures are consistently reported by researchers, enabling community - wide conclusions to be drawn ;
  3. They must be computationally tractable, to match the rapid growth in the scale of modern data collection.
  4. They must be simple to report as a single number for each method - dataset combination.
- Performance metrics for binary classification are designed to capture trade-offs between four fundamental population quantities : True positives, false positives, true negatives and false negatives.

- The evaluation measures in classification problems are defined from a matrix with the numbers of examples correctly and incorrectly classified for each class, named confusion matrix.
- Confusion matrix is also called a contingency table.
  1. **False positives** : Examples predicted as positive, which are from the negative class.
  2. **False negatives** : Examples predicted as negative, whose true class is positive.
  3. **True positives** : Examples correctly predicted as pertaining to the positive class.
  4. **True negatives** : Examples correctly predicted as belonging to the negative class.

$$\text{Accuracy rate} = \frac{|\text{True negatives}| + |\text{True positives}|}{|\text{False negatives}| + |\text{True positive}| + |\text{True negatives}| + |\text{True positives}|}$$

- The complement of accuracy rate is the error rate, which evaluates a classifier by its percentage of incorrect predictions.

$$\text{Error rate} = \frac{|\text{False negatives}| + |\text{False positives}|}{|\text{False negatives}| + |\text{False positive}| + |\text{True negatives}| + |\text{True positives}|}$$

$$\text{Error rate} = 1 - (\text{Accuracy rate})$$

- The recall and specificity measures evaluate the effectiveness of a classifier for each class in the binary problem. The recall is also known as sensitivity or true positive rate. Recall is the proportion of examples belonging to the positive class which were correctly predicted as positive.
- The **specificity** is a statistical measure of how well a binary classification test correctly identifies the negatives cases.

$$\text{Recall (R)} = \frac{|\text{True positive}|}{|\text{True positive}| + |\text{False negative}|}$$

$$\text{Specificity} = \frac{|\text{True negatives}|}{|\text{False positives}| + |\text{True negative}|}$$

- True Positive Rate (TPR) is also called sensitivity, hit rate, and recall.

$$\text{Sensitivity} = \frac{\text{Number of true positives}}{\text{Number of true positives} + \text{Number of false positives}}$$

- A statistical measure of how well a binary classification test correctly identifies a conditon. Probability of correctly labeling members of the target class.

- No single measure tells the whole story. A classifier with 90 % accuracy can be useless if 90 percent of the population does not have cancer and the 10 % that do are misclassified by the classifier. Use of multiple measures recommended.
- Binary classification accuracy metrics quantify the two types of correct predictions and two types of errors. Typical metrics are accuracy (ACC), precision, recall, false positive rate, F1-measure. Each metric measures a different aspect of the predictive model.

### 5.6.1 Issues Regarding Classification and Prediction

#### Preparing the data for classification and prediction

Following pre-processing steps may be applied to the data to help improve the accuracy, efficiency, and scalability of the classification or prediction process.

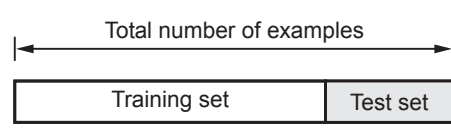
- a) **Data cleaning** : Preprocess data in order to reduce noise and handle missing values.
- b) **Relevance analysis (Feature selection)** : Remove the irrelevant or redundant attributes.
- c) **Data transformation** : Generalize and/or normalize data

Feature wise comparison between classification and prediction :

1. **Accuracy** : Accuracy of classifier refers to the ability of classifier. It predict the class label correctly and the accuracy of the predictor refers to how well a given predictor can guess the value of predicted attribute for a new data.
2. **Speed** : This refers to the computational cost in generating and using the classifier or predictor.
3. **Robustness** : It refers to the ability of classifier or predictor to make correct predictions from given noisy data.
4. **Scalability** : Scalability refers to the ability to construct the classifier or predictor efficiently ; given large amount of data.
5. **Interpretability** : It refers to what extent the classifier or predictor understands.

### 5.6.2 Holdout Method

- The data is split into two different datasets labelled as a training and a testing dataset. This can be a 60/40 or 70/30 or 80/20 split. This technique is called the hold-out validation technique.
- Suppose we have a database with house prices as the dependent variable and two independent variables showing the square footage of the house and the number of rooms.

- Now, imagine this dataset has 30 rows. The whole idea is that you build a model that can predict house prices accurately.
  - To 'train' your model, or see how well it performs, we randomly subset 20 of those rows and fit the model.
  - The second step is to predict the values of those 10 rows that we excluded and measure how well our predictions were.
  - As a rule of thumb, experts suggest to randomly sample 80 % of the data into the training set and 20 % into the test set.
- 
- Fig. 5.6.1**
- Training set : Used to train the classifier.
  - The holdout method has two, basic drawbacks :
    1. It requires extra dataset
    2. It is a single train-and-test experiment, the holdout estimate of error rate will be misleading if we happen to get an "unfortunate" split.

### 5.6.3 Random Subsampling

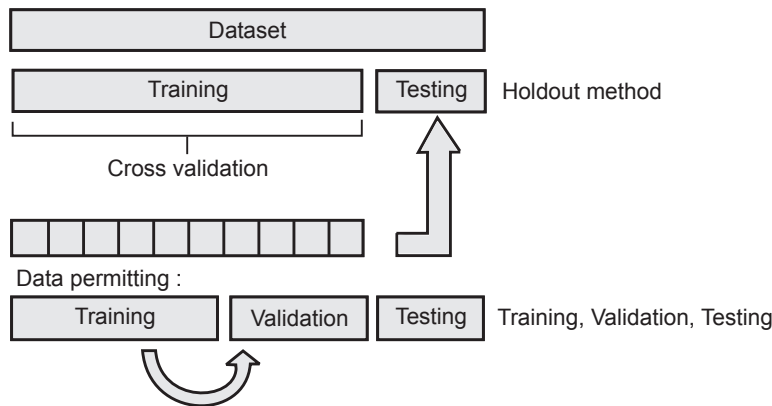
- Random subsampling performs K data splits of the entire sample. For each data split, a fixed number of observations is chosen without replacement from the sample and kept aside as the test data.
- The prediction model is fitted to the training data from scratch for each of the K splits and an estimate of the prediction error is obtained from each test set.
- Let the estimated PE in the  $i^{\text{th}}$  test set be denoted by  $E_i$ . The true error estimate is obtained as the average of the separate estimates  $E_i$ .

$$\frac{1}{K} \sum_{i=1}^K E_i$$

#### 1. Cross-Validation

- Cross-validation is a technique for evaluating estimating performance by training several machine learning models on subsets of the available input data and evaluating them on the complementary subset of the data. Use cross-validation to detect overfitting, i.e., failing to generalize a pattern.
- In general, machine learning involves deriving models from data, with the aim of achieving some kind of desired behaviour, e.g., prediction or classification.

- Fig. 5.6.2 shows cross-validation.

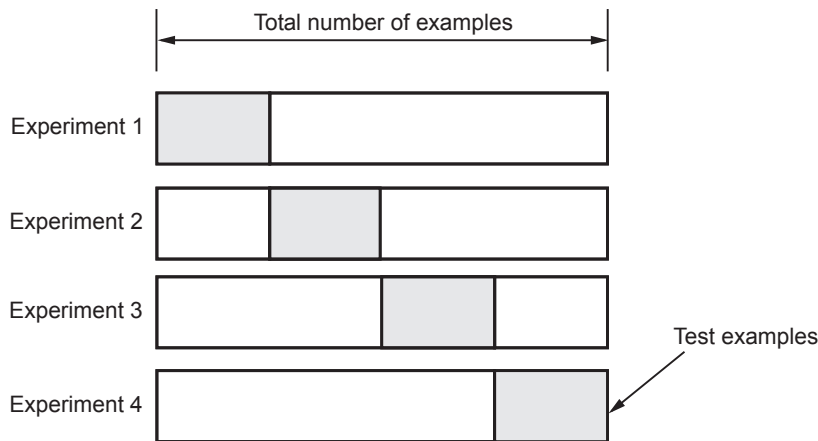


**Fig. 5.6.2 Cross validation**

- But this generic task is broken down into a number of special cases. When training is done, the data that was removed can be used to test the performance of the learned model on "new" data. This is the basic idea for a whole class of model evaluation methods called **cross validation**.
- Types of cross validation methods are holdout, K-fold and Leave-one-out.
- The holdout method is the simplest kind of cross validation. The data set is separated into two sets, called the training set and the testing set. The function approximate fits a function using the training set only.
- The K-fold cross validation is one way to improve over the holdout method. The data set is divided into k subsets, and the holdout method is repeated k times.
- Each time, one of the k subsets is used as the test set and the other k - 1 subsets are put together to form a training set. Then the average error across all k trials is computed.
- Leave-one-out cross validation is K-fold cross validation taken to its logical extreme, with K equal to N, the number of data points in the set.
- That means that N separate times, the function approximate is trained on all the data except for one point and a prediction is made for that point.
- Cross-validation ensures non-overlapping test sets.

#### **K-fold cross-validation :**

- In this technique, k - 1 folds are used for training and the remaining one is used for testing as shown in Fig. 5.6.3.



**Fig. 5.6.3 K-fold cross validation**

- The advantage is that entire data is used for training and testing. The error rate of the model is average of the error rate of each iteration.
- This technique can also be called a form the repeated hold-out method. The error rate could be improved by using stratification technique.

### Review Question

1. Explain any three of classification performance measures.

**SPPU : Dec.-18 (End Sem), Marks 6**

## 5.7 Clustering and Time-series Analysis using Scikit-learn

- Time series data is widely used to analyse different trends and seasonalities of products over time by various industries. Sktime is a unified python framework/library providing API for machine learning with time series data and sklearn compatible tools to analyse, visualize, tune and validate multiple time series learning models such as time series forecasting, time series regression and classification.
- Time series are a stream of data that are created by making measures of something such as sales, temperature, stocks, etc. in fixed frequency. They have to be indexed in time order and usually used in weather forecasting, econometrics, earthquake prediction, signal processing, etc.
- Clustering of unlabeled data can be performed with the module sklearn.cluster.

### 5.7.1 Scikit-learn

- Scikit-learn is a free machine learning library for Python. It features various algorithms like support vector machine, random forests, and k-neighbours, and it also supports Python numerical and scientific libraries like NumPy and SciPy.

- It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use.
- Scikit-learn is a library, i.e. a collection of classes and functions that users import into Python programs. Using scikit-learn therefore requires basic Python programming knowledge
- The library is built upon the SciPy (Scientific Python) that must be installed before you can use scikit-learn. This stack that includes :
  1. NumPy : Base n-dimensional array package
  2. SciPy : Fundamental library for scientific computing
  3. Matplotlib : Comprehensive 2D/3D plotting
  4. IPython : Enhanced interactive console
  5. SymPy : Symbolic mathematics
  6. Pandas : Data structures and analysis
- Scikit-learn is the package for machine learning and data science experimentation favoured by most data scientists. It contains a wide range of well-established learning algorithms, error functions, and testing procedures.

## 5.7.2 Understanding Classes in Scikit-learn

- Scikit-learn takes a highly object-oriented approach to machine learning models. Every major Scikit-learn class inherits from *sklearn.base.BaseEstimator*.
- Scikit-learn features some base classes on which all the algorithms are built. Apart from *BaseEstimator*, the class from which all other classes inherit, there are four class types covering all the basic machine learning functionalities :
  1. Classifying
  2. Regressing
  3. Grouping by clusters
  4. Transforming data
- Scikit-learn takes a highly object-oriented approach to machine learning models. Every major Scikit-learn class inherits from *sklearn.base.BaseEstimator*.
- All objects within scikit-learn share a uniform common basic API consisting of three complementary interfaces : an **estimator** interface for building and fitting models, a **predictor** interface for making predictions and a **transformer** interface for converting data.

### 1. Estimators

- The estimator interface is at the core of the library. It defines instantiation mechanisms of objects and exposes a fit method for learning a model from training data.

- All supervised and unsupervised learning algorithms (e.g., for classification, regression or clustering) are offered as objects implementing this interface. Machine learning tasks like feature extraction, feature selection or dimensionality reduction are also provided as estimators.

## 2. Predictors

- The predictor interface extends the notion of an estimator by adding a predict method that takes an array  $X$  test and produces predictions for  $X$  test, based on the learned parameters of the estimator.
- In the case of supervised learning estimators, this method typically returns the predicted labels or values computed by the model.

## 3. Transformers

- Since it is common to modify or filter data before feeding it to a learning algorithm, some estimators in the library implement a transformer interface which defines a transform method.
- It takes as input some new data  $X$  test and yields as output a transformed version of  $X$  test.
- Preprocessing, feature selection, feature extraction and dimensionality reduction algorithms are all provided as transformers within the library.

## 5.8 Confusion Matrix

**SPPU : May-19**

- A confusion matrix is a table that is often used to describe the performance of a classification model on a set of test data for which the true values are known. The numbers displayed give the frequency of each data point.
- The confusion matrix for binary classification shown below :

|            | Predicted class |                |
|------------|-----------------|----------------|
| True class | Positive        | Negative       |
| Positive   | True negative   | False negative |
| Negative   | False positive  | True negative  |

- A confusion matrix contains information about actual and predicted classification done by a classification system. Performance of such systems is commonly evaluated using the data in the matrix. Confusion matrix is also called a contingency table.
1. **False positives** : Examples predicted as positive, which are from the negative class.

2. **False negatives** : Examples predicted as negative, whose true class is positive.
  3. **True positives** : Examples correctly predicted as pertaining to the positive class.
  4. **True negatives** : Examples correctly predicted as belonging to the negative class.
- Binary classification accuracy metrics quantify the two types of correct predictions and two types of errors. Typical metrics are accuracy (ACC), precision, recall, false positive rate, F1-measure. Each metric measures a different aspect of the predictive model.
  - Accuracy (ACC) measures the fraction of correct predictions. Precision measures the fraction of actual positives among those examples that are predicted as positive. Recall measures how many actual positives were predicted as positive. F1-measure is the harmonic mean of precision and recall many actual positives were predicted as positive. F1-measure is the harmonic mean of precision and recall.

### 5.8.1 ROC Curve

- Receiver Operating Characteristics (ROC) graphs have long been used in signal detection theory to depict the tradeoff between hit rates and false alarm rates noisy channel. Recent years have seen an increase in the use of ROC graphs in the machine learning community.
- An ROC plot plots true positive rate on the Y-axis against false positive rate on the X-axis ; a single contingency table corresponds to a single point in an ROC plot.
- The performance of a ranker can be assessed by drawing piecewise linear curve in an ROC plot, known as an ROC curve. The curve starts in (0, 0), finishes in (1, 1) and is monotonically non-decreasing in both axes.
- A useful technique for organizing classifiers and visualizing their performance. Especially useful for domains with skewed class distribution and unequal classification error costs.
- It allows to create ROC curve and a complete sensitivity / specificity report. The ROC curve is a fundamental tool for diagnostic test evaluation.
- In a ROC curve the true positive rate (sensitivity) is plotted in function of the false positive rate (100 -specificity) for different cut-off points of a parameter. Each point on the ROC curve represents a sensitivity / specificity pair corresponding to a particular decision threshold. The area under the ROC curve is a measure of how well a parameter two segments.

- Each point on an ROC curve connecting two segments corresponds to the true and false positive rates achieved on the same test set by the classifier obtained from the ranker by splitting the ranking between those two segments.
- An ROC curve is convex if the slopes are monotonically non-increasing when moving along the curve from (0, 0) to (1, 1). A concavity in an ROC curve, i.e. in an ROC curve, i.e. two or more adjacent segment with increasing slopes, indicates a locally worse than random ranking. In this case, we would get better ranking performance by joining the segments involved in concavity, thus creating a coarser classifier.

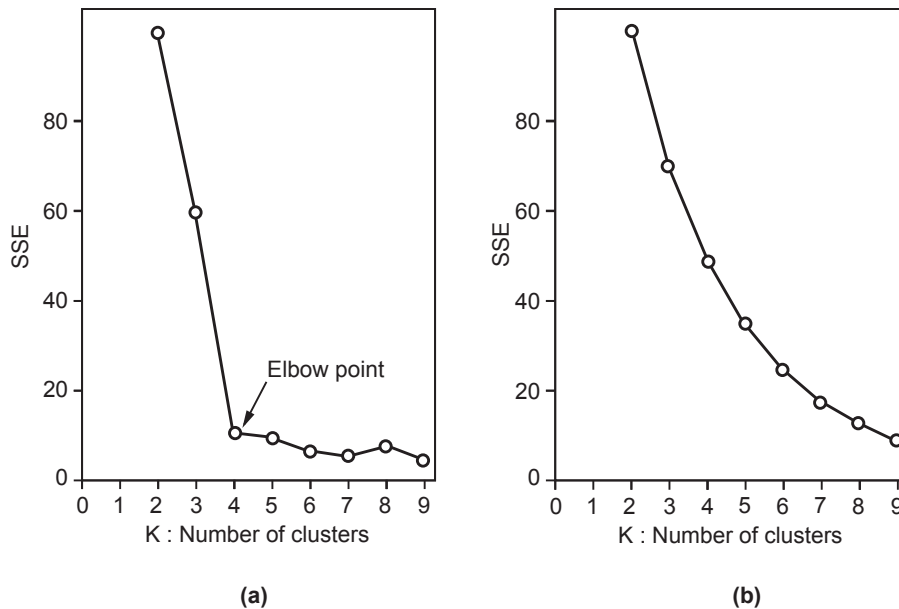
### Review Question

1. Explain the term confusion matrix.

**SPPU : May-19 (End Sem), Marks 4**

## 5.9 Elbow Plot

- The elbow method is used to determine the optimal number of clusters in k-means clustering. The elbow method plots the value of the cost function produced by different values of k.
- If k increases, average distortion will decrease, each cluster will have fewer constituent instances, and the instances will be closer to their respective centroids. However, the improvements in average distortion will decline as k increases. The value of k at which improvement in distortion declines the most is called the elbow, at which we should stop dividing the data into further clusters.
- It involves running the algorithm multiple times over a loop, with an increasing number of cluster choice and then plotting a clustering score as a function of the number of clusters.
- The Elbow and Silhouette methods are the two state-of-the-art methods used to identify the correct cluster number in the dataset.
- The Elbow method is the oldest method to distinguish the potential optimal cluster number for the analyzed dataset, whose basic idea is to specify  $K = 2$  as the initial optimal cluster number  $K$ , and then keeps increasing  $K$  by step 1 to the maximal specified for the estimated potential optimal cluster number, and finally distinguish the potential optimal cluster number  $K$  corresponding to the plateau.
- The optimal cluster number  $K$  is distinguished by the fact that before reaching  $K$ , the cost rapidly decreases to the called cost peak value, and after exceeding  $K$ , it continues to increase with the called cost peak value almost unchanged, as shown in Fig. 5.9.1 (a) with an explicit elbow point.



**Fig. 5.9.1 Elbow point**

- Meanwhile, the optimal cluster number corresponding to the elbow point depends on the manmade selection. There is, however, a problem with the Elbow method in that the elbow point cannot be unambiguously distinguished by the experienced analysts when the plotted curve is fairly smooth, as shown in Fig. 5.9.1 (b) with an ambiguous elbow point.
- To select the best K, we need to plot the mean in-cluster distance for each K. As K increases from 1, before reaching the optimal K, the decrease speed is relatively fast because the number of centers are too low from the very beginning and each new center will incur a large decrease in the mean distance.
- But after the optimal K, the decrease is slow since the correct cluster structure is already discovered and any newly added center will appear in a certain cluster already formed. That will not decrease the mean in-cluster distance too much. The entire curve looks like an L shape and the best K lies in the turning point or the elbow of the L shape.

## 5.10 Multiple Choice Questions

**Q.1** A \_\_\_\_\_ is a flowchart-like tree structure, where each internal node denotes a test on an attribute.

- |                                          |                                          |
|------------------------------------------|------------------------------------------|
| <input type="checkbox"/> a decision tree | <input type="checkbox"/> b binary tree   |
| <input type="checkbox"/> c cluster       | <input type="checkbox"/> d none of these |

**Q.2 A node without further branches is called as \_\_\_\_.**

- |                                          |                                        |
|------------------------------------------|----------------------------------------|
| <input type="checkbox"/> a internal node | <input type="checkbox"/> b root node   |
| <input type="checkbox"/> c lead node     | <input type="checkbox"/> d binary node |

**Q.3 \_\_\_\_\_ occurs when the gap between the training error and test error is too large.**

- |                                         |                                        |
|-----------------------------------------|----------------------------------------|
| <input type="checkbox"/> a Underfitting | <input type="checkbox"/> b Overfitting |
| <input type="checkbox"/> c Overloaded   | <input type="checkbox"/> d Purning     |

**Q.4 What is the approach of basic algorithm for decision tree induction ?**

- |                                       |                                               |
|---------------------------------------|-----------------------------------------------|
| <input type="checkbox"/> a Greedy     | <input type="checkbox"/> b Top down           |
| <input type="checkbox"/> c Procedural | <input type="checkbox"/> d Step by Procedural |

**Q.5 What are two steps of tree pruning work ?**

- |                                                                               |
|-------------------------------------------------------------------------------|
| <input type="checkbox"/> a Pessimistic pruning and Optimistic pruning         |
| <input type="checkbox"/> b Postpruning and Prepruning                         |
| <input type="checkbox"/> c Cost complxity pruning and time complexity pruning |
| <input type="checkbox"/> d None of the options                                |

**Q.6 How will you counter over-fitting in decision tree ?**

- |                                                                                        |
|----------------------------------------------------------------------------------------|
| <input type="checkbox"/> a By pruning the longer rules                                 |
| <input type="checkbox"/> b By creating new rules                                       |
| <input type="checkbox"/> c Both by pruning the longer rules and by creating new rules. |
| <input type="checkbox"/> d None of the above                                           |

**Q.7 Which of the following is not a forecasting technique ?**

- |                                         |                                        |
|-----------------------------------------|----------------------------------------|
| <input type="checkbox"/> a Judgemental  | <input type="checkbox"/> b Time series |
| <input type="checkbox"/> c Time horizon | <input type="checkbox"/> d Associative |

**Q.8 Which of the following is not true for forecasting ?**

- |                                                                                               |
|-----------------------------------------------------------------------------------------------|
| <input type="checkbox"/> a Forecasts are rarely perfect.                                      |
| <input type="checkbox"/> b The underlying casual system will remain same in the future.       |
| <input type="checkbox"/> c Forecast for group of items is accurate than individual item.      |
| <input type="checkbox"/> d Short range forecasts are less accurate than long range forecasts. |

**Q.9 ETL stand for \_\_\_\_\_.**

- ☐ a Extract Transform and Load
- ☐ b Exact Transfer and Language
- ☐ c Extract Transmission and Language
- ☐ d None

**Q.10 ARIMA is a \_\_\_\_\_ model that uses time series data to either better understand the data set or to predict future trends.**

- ☐ a statistical analysis
- ☐ b analytical
- ☐ c descriptive
- ☐ d all of these

**Answer Keys for Multiple Choice Questions :**

|     |   |     |   |     |   |     |   |      |   |
|-----|---|-----|---|-----|---|-----|---|------|---|
| Q.1 | a | Q.2 | c | Q.3 | b | Q.4 | a | Q.5  | b |
| Q.6 | a | Q.7 | c | Q.8 | d | Q.9 | a | Q.10 | a |

